

```
//ESP32

#include <SPI.h>

#include <WiFi.h>

#include <ArduinoJson.h>

#include <ThingyESP32.h>

#include <Wire.h>

#include <Adafruit_ADS1X15.h>

#include <WiFiUdp.h>

#include <NTPClient.h>

#include <TimeLib.h>

#include "CTBot.h"

#include "Utilities.h"


#define _DEBUG_

#define CTBOT_DEBUG_MODE

#define CTBOT_DEBUG_ALL

#define THINGER_SERIAL_DEBUG

#define USERNAME "smandini"

#define DEVICE_ID "undergraduated"

#define DEVICE_CREDENTIAL "undergraduated88"


//Pin sensor flow

#define flowPin1 5

#define flowPin2 18

#define flowPin3 33

//WiFi
```

```
#define SSID "Sofrosine"

#define SSID_PASSWORD "weganggang"

//NTP

const long utcOffsetInSeconds = 28800; // set offset


WiFiUDP ntpUDP;

NTPClient timeClient(ntpUDP, "id.pool.ntp.org", utcOffsetInSeconds);


char Time[ ] = "Jam: 00:00:00";
char Date[ ] = "Tgl: 00/00/2000";
byte last_second, second_, minute_, hour_, day_, month_;
int year_;


ThingyESP32 thingy(USERNAME, DEVICE_ID, DEVICE_CREDENTIAL);

//Inisialisasi Thingy


WiFiClientSecure client;


CTBot myBot; //variabel untuk bot Telegram

String token = "6400864545:AAHNAQWa-3IHwnF839P1ZMiyHATQBTX3Mq4";

int64_t CHAT_ID = -1002071693755;


Adafruit_ADS1115 ads; //Inisialisasi ADC ADS1115


//Variabel Global Sensor Tekanan

int16_t adc0, adc1, adc2, adc3;
```

```

float volts0, volts1, volts2, volts3;

float pressure0, pressure1, pressure2, pressure3;


float baselineVoltage0;

float baselineVoltage1;

float baselineVoltage2;

float baselineVoltage3;


// Variabel global untuk sensor flow

volatile int pulseCount1; // Jumlah pulse dari sensor

volatile int pulseCount2; // Jumlah pulse dari sensor


volatile int pulseCount3; // Jumlah pulse dari sensor

float flowRate1;

float flowRate2;

float flowRate3; //Debit air dalam liter per menit yang dihasilkan pembacaan sensor

unsigned long oldTime; // Waktu terakhir ketika pulse terdeteksi


float A, B, C, D;


String currentKondisi = "";

String currentLokasi = "";


// Max 2 Desimal

float round_to_dp( float in_value, int decimal_place )

{

```

```
float multiplier = powf( 10.0f, decimal_place );  
in_value = roundf( in_value * multiplier ) / multiplier;  
return in_value;  
}
```

```
// Variabel untuk kalibrasi sensor flow
```

```
// 5,5 berasal dari datasheet sensor
```

```
// Pulse frequency (Hz) in Horizontal Test= 5.5Q, Q is flow rate in L/min
```

```
float calibrationFactor1 = 5.5; // Faktor kalibrasi sensor 1
```

```
float calibrationFactor2 = 5.5; // Faktor Kalibrasi sensor 2
```

```
float calibrationFactor3 = 5.5; //Faktor Kalibrasi sensor 3
```

```
void ICACHE_RAM_ATTR pulseCounter1();
```

```
void ICACHE_RAM_ATTR pulseCounter2();
```

```
void ICACHE_RAM_ATTR pulseCounter3();
```

```
unsigned long waktutelegram1 = 0;
```

```
unsigned long waktutelegram2 = 0;
```

```
void setup() {
```

```
// Mengaktifkan serial communication
```

```
Serial.begin(115200);
```

```
connectToWiFi();
```

```
//Inisialisasi NTP
```

```
timeClient.begin();
```

```
// pinMode(BUILTIN_LED, OUTPUT);  
thing.add_wifi(SSID, SSID_PASSWORD);  
  
//ADS  
if (!ads.begin())  
{  
  Serial.println("Failed to initialize ADS.");  
  while (1);  
  yield();  
}  
  
// (Inisialisasi) Mengatur pin sensor water flow sebagai input dan mengaktifkan pull-up  
resistor internal  
pinMode(flowPin1, INPUT_PULLUP);  
pinMode(flowPin2, INPUT_PULLUP);  
pinMode(flowPin3, INPUT_PULLUP);  
// Reset semua variabel  
pulseCount1 = 0; pulseCount2 = 0; pulseCount3 = 0;  
flowRate1 = 0.0; flowRate2 = 0.0; flowRate3 = 0.0;  
oldTime = 0;  
  
// Menginisialisasi interrupt dgn 3 argumen ke pin sensor  
attachInterrupt(digitalPinToInterrupt(flowPin1), pulseCounter1, FALLING);  
attachInterrupt(digitalPinToInterrupt(flowPin2), pulseCounter2, FALLING);  
attachInterrupt(digitalPinToInterrupt(flowPin3), pulseCounter3, FALLING);
```

```
thing["millis"] >> outputValue(millis());
```

```
//OUTPUT dari sensor menggunakan operator >>
```

```
thing["sensor"] >> [(pson & out){  
  out["Waterflow Sensor 1"] = round_to_dp( flowRate1, 2);  
  out["Waterflow Sensor 2"] = round_to_dp( flowRate2, 2);  
  out["Waterflow Sensor 3"] = round_to_dp( flowRate3, 2);  
  out["Pressure Sensor 1"] = round_to_dp( pressure1, 2);  
  out["Pressure Sensor 2"] = round_to_dp( pressure2, 2);  
  out["Pressure Sensor 3"] = round_to_dp( pressure3, 2);  
  out["Kondisi"] = String(currentKondisi);  
  out["Lokasi"] = String(currentLokasi);  
};
```

```
// output resource (Kondisi)
```

```
thing["Kondisi"] >> outputValue(currentKondisi);
```

```
// output resource (Lokasi)
```

```
thing["Lokasi"] >> outputValue(currentLokasi);
```

```
}
```

```
void loop() {
```

```
//Memeriksa terhubung ke WiFi
```

```
if (WiFi.status() != WL_CONNECTED) {
```

```
  Serial.println("Koneksi Wi-Fi terputus. Mencoba untuk terhubung kembali...");
```

```
  reconnectToWiFi();
```

```
}
```

```
thing.handle();
```

```
//Perbarui waktu dari NTP
```

```
timeClient.update();
```

```
unsigned long unix_epoch = timeClient.getEpochTime(); // Unix epoch time dari NTP  
server
```

```
second_ = second(unix_epoch);
```

```
if (last_second != second_) {
```

```
minute_ = minute(unix_epoch);
```

```
hour_ = hour(unix_epoch);
```

```
day_ = day(unix_epoch);
```

```
month_ = month(unix_epoch);
```

```
year_ = year(unix_epoch);
```

```
Time[12] = second_ % 10 + 48;
```

```
Time[11] = second_ / 10 + 48;
```

```
Time[9] = minute_ % 10 + 48;
```

```
Time[8] = minute_ / 10 + 48;
```

```
Time[6] = hour_ % 10 + 48;
```

```
Time[5] = hour_ / 10 + 48;
```

```
Date[5] = day_ / 10 + 48;
```

```
Date[6] = day_ % 10 + 48;  
Date[8] = month_ / 10 + 48;  
Date[9] = month_ % 10 + 48;  
Date[13] = (year_ / 10) % 10 + 48;  
Date[14] = year_ % 10 % 10 + 48;
```

```
if((millis() - oldTime) >= 1000 * 1) //setiap 1 detik
```

```
{
```

```
detachInterrupt(sensorInterrupt); //Nonaktif interrupt
```

```
// Karena interupsi telah dimatikan, fungsi millis() tidak akan terus menghitung waktu  
secara real-time
```

```
// Namun, fungsi ini masih akan mengembalikan nilai waktu terakhir yang telah dicatat
```

```
// sebelum interupsi dimatikan
```

```
// Jadi, meskipun waktu tidak bertambah saat interupsi dimatikan
```

```
// kita masih bisa mendapatkan nilai waktu sebelum itu
```

```
unsigned long currentTime = millis();
```

```
// Menghitung selang waktu sejak deteksi pulse terakhir
```

```
unsigned long elapsedTime = currentTime - oldTime;
```

```
// Mengupdate waktu terakhir ketika pulse terdeteksi
```

```
oldTime = currentTime;
```

```
// Menghitung debit air dalam liter per menit
```

```
// Untuk mengubah milidetik menjadi detik, akan dibagi menjadi 1000 (1s = 1000ms)
```

```
flowRate1 = (((1000.0 / elapsedTime) * pulseCount1) / calibrationFactor1);  
flowRate2 = (((1000.0 / elapsedTime) * pulseCount2) / calibrationFactor2);  
flowRate3 = (((1000.0 / elapsedTime) * pulseCount3) / calibrationFactor3);
```

```
pulseCount1 = 0; pulseCount2 = 0; pulseCount3 = 0;
```

```
pressureSensor();
```

```
receiveMessage();
```

```
checkLeakage();
```

```
// dari update vy device
```

```
// streaming resources data
```

```
if (millis() % (60 * 1000) == 0) {
```

```
thing.stream(thing["sensor"]);
```

```
thing.write_bucket("underthesis2", "sensor");
```

```
thing.stream("sensor");
```

```
oldTime = millis();
```

```
}
```

```
// Serial.print(" Voltage 1: ");
```

```
// Serial.println(volts0, 3);
```

```
Serial.print(" Voltage 2: ");
```

```
Serial.println(volts1, 3);
```

```
// Serial.print(" Voltage 3: ");
// Serial.println(volts2, 3);
// Serial.print(" Voltage 4: ");
// Serial.println(volts3, 3);
// Serial.print(" Tekanan1: ");
// Serial.print(pressure1);
// Serial.print(" PSI");
// Serial.print(" Flow 1: ");
// Serial.println(flowRate1, 3);
// Serial.print("Voltage 2: ");
// Serial.print(volts1);
// Serial.print(" Tekanan2: ");
// Serial.print(pressure2);
// Serial.print(" PSI");
// Serial.print(" Flow 2: ");
// Serial.println(flowRate2, 3);
// Serial.print("Voltage 3: ");
// Serial.print(volts3);
// Serial.print(" Tekanan3: ");
// Serial.print(pressure3);
// Serial.print(" PSI");
// Serial.print(" Flow 3: ");
// Serial.println(flowRate3, 3);
```

```
// Serial.print(" Status: ");  
// Serial.print(currentKondisi);  
// Serial.print(" Lokasi: ");  
// Serial.println(currentLokasi);  
}  
}  
}
```

```
void pulseCounter1()  
{  
// Meningkatkan jumlah pulse  
pulseCount1++;  
}
```

```
void pulseCounter2() {  
// Meningkatkan jumlah pulse  
pulseCount2++;  
}
```

```
void pulseCounter3() {  
// Meningkatkan jumlah pulse  
pulseCount3++;  
}
```

```
void connectToWiFi() {  
Serial.print("Memulai telegram bot. Mencoba untuk terhubung ke Wi-Fi");
```

```
myBot.wifiConnect(SSID, SSID_PASSWORD);

//telegram token
myBot.setTelegramToken(token);


int percobaan = 0;


while (WiFi.status() != WL_CONNECTED && percobaan < 10) {
yield();
Serial.print(".");
percobaan++;
}


if (WiFi.status() == WL_CONNECTED) {
Serial.println("\nBerhasil Terhubung");
Serial.print("IP address : ");
Serial.println(WiFi.localIP());
if(myBot.sendMessage(CHAT_ID, "Bot Monitoring Water Leak has
started\n/informasi\n")) {
Serial.println("Pesan Berhasil Terkirim");
} else {
Serial.println("Pesan Tidak Terkirim");
}
} else {
Serial.println("\nGagal terhubung");
}
}
```

```
void reconnectToWiFi() {  
    Serial.println("Mencoba restart koneksi Wi-Fi...");  
    // Hentikan koneksi Wi-Fi  
    WiFi.disconnect();  
    // Tunggu sebentar sebelum mencoba untuk terhubung kembali  
    delay(2000);  
    // Coba untuk terhubung kembali  
    connectToWiFi();  
}  
  
void receiveMessage() {  
  
    TBMessage msg;  
  
    if (myBot.getNewMessage(msg)) {  
  
        String isipesan = msg.text;  
  
        // check if the message is a text message  
  
        // if (msg.messageType == CTBotMessageText) {  
  
        if (isipesan == "/informasi@monitoringwaterleak_bot") {  
            String pesan = "Welcome, " + msg.sender.firstName + " " + msg.sender.lastName  
            + ".\n";
```

```
pesan += "Monitoring Kebocoran: \n";  
pesan += String(Date) + "\n";  
pesan += String(Time) + "\n";  
pesan += "Kondisi Kebocoran: " + String(currentKondisi) + "\n";  
pesan += "Lokasi Kebocoran: " + String(currentLokasi) + "\n";  
myBot.sendMessage(CHAT_ID, pesan, "");
```

```
Serial.print("Mengirim pesan ke telegram...");  
}
```

```
if (isipesan != "/informasi@monitoringwaterleak_bot") {  
String pesanmsg1 = "Welcome, " + msg.sender.firstName + " " +  
msg.sender.lastName + ".\n";  
pesanmsg1 += "Gunakan perintah berikut untuk mendapatkan informasi: \n";  
pesanmsg1 += "/informasi";  
myBot.sendMessage(CHAT_ID, pesanmsg1, "");
```

```
Serial.print("Mengirim pesan ke telegram...");  
}
```

```
if (isipesan != "/informasi@monitoringwaterleak_bot" && currentKondisi ==  
"AREA 1 & 2") {
```

```
String pesan3 = "PERINGATAN! \n";  
pesan3 += "Tanggal: " + String(Date) + "\n";  
pesan3 += "Waktu: " + String(Time) + "\n";
```

```
pesan3 += "Kondisi Kebocoran: " + currentKondisi + "\n";  
pesan3 += "Lokasi Kebocoran: " + currentLokasi + "\n";  
myBot.sendMessage(CHAT_ID, pesan3, "");  
}
```

```
delay(50);  
}  
}
```

```
void pressureSensor() {  
  //Baca nilai ADC dari saluran 0-3  
  adc0 = ads.readADC_SingleEnded(0);  
  adc1 = ads.readADC_SingleEnded(1);  
  adc2 = ads.readADC_SingleEnded(2);  
  adc3 = ads.readADC_SingleEnded(3);
```

```
  // Konversi pembacaan nilai analog ke digital (ADC)
```

```
  // ADS1115 memiliki resolusi 16-bit (32767)
```

```
  volts0 = ads.computeVolts(adc0);
```

```
  volts1 = ads.computeVolts(adc1);
```

```
  volts2 = ads.computeVolts(adc2);
```

```
  volts3 = ads.computeVolts(adc3);
```

```
  // Konversi tegangan ke tekanan untuk pembacaan tekanan fluida(dalam satuan PSI)
```

```
  // baselineVoltage adalah tegangan keluaran sensor yang terukur tanpa adanya beban
```

```
  // 174 merupakan jumlah max beban tekanan pada sensor (dalam satuan PSI)
```

```
// 4,5 merupakan jumlah max tegangan keluaran

// pressure0 = (volts0 - baselineVoltage0) * (174.0 / (4.5 - baselineVoltage0));
pressure1 = (volts1 - baselineVoltage1) * (174.0 / (4.5 - baselineVoltage1));

pressure2 = (volts2 - baselineVoltage2) * (174.0 / (4.5 - baselineVoltage2));
pressure3 = (volts3 - baselineVoltage3) * (174.0 / (4.5 - baselineVoltage3));


// output sensor tanpa beban

//baselineVoltage0 = 0.51;

baselineVoltage1 = 0.59000; //at zero pressure

baselineVoltage2 = 0.41999;

baselineVoltage3 = 0.41878;


if (pressure1 < 0) {
    pressure1 = 0; // Set nilai P menjadi 0 jika negatif
}

if (pressure2 < 0) {
    pressure2 = 0; // Set nilai P menjadi 0 jika negatif
}

if (pressure3 < 0) {
    pressure3 = 0; // Set nilai P menjadi 0 jika negatif
}
}
```

```

void checkLeakage() {

    unsigned long currentTime = millis();

    // Menghitung selang waktu sejak deteksi pulse terakhir
    unsigned long elapsedTime = currentTime - oldTime;

    // Mengupdate waktu terakhir ketika pulse terdeteksi
    oldTime = currentTime;

    // Absolute
    A = abs(flowRate2 - flowRate1);
    B = abs(flowRate3 - flowRate2);
    C = 0.04 * flowRate1;
    D = 0.04 * flowRate2;

    // Periksa kebocoran di Area 1
    if (A > C) {
        currentKondisi = "BOCOR";
        thing.stream(thing["Kondisi"]);
        currentLokasi = "AREA 1";
        thing.stream(thing["Lokasi"]);

        if (currentTime - waktutelegram1 >= 300000) { // 300000 ms = 5 minutes
            String pesan1 = "PERINGATAN! \n";
            pesan1 += "Tanggal: " + String(Date) + "\n"; // Pastikan Date terdefinisi atau ganti

```

dengan fungsi yang sesuai

```
pesan1 += "Waktu: " + String(Time) + "\n"; // Pastikan Time terdefinisi atau ganti
```

dengan fungsi yang sesuai

```
pesan1 += "Kondisi Kebocoran: " + currentKondisi + "\n";
```

```
pesan1 += "Lokasi Kebocoran: " + currentLokasi + "\n";
```

```
myBot.sendMessage(CHAT_ID, pesan1, "");
```

```
waktutelegram1 = millis();
```

```
}
```

```
// Periksa kebocoran di Area 2
```

```
if (B > D) {
```

```
currentKondisi = "BOCOR";
```

```
thing.stream(thing["Kondisi"]);
```

```
currentLokasi = "AREA 1 & 2";
```

```
thing.stream(thing["Lokasi"]);
```

```
if (currentTime - waktutelegram2 >= 300000) { // 300000 ms = 5 minutes
```

```
String pesan4 = "PERINGATAN! \n";
```

```
pesan4 += "Tanggal: " + String(Date) + "\n";
```

```
pesan4 += "Waktu: " + String(Time) + "\n";
```

```
pesan4 += "Kondisi Kebocoran: " + currentKondisi + "\n";
```

```
pesan4 += "Lokasi Kebocoran: AREA 2 \n";
```

```
myBot.sendMessage(CHAT_ID, pesan4, "");
```

```
waktutelegram2 = millis();
```

```
}
```

```
}
```

```
} else {  
    if (B > D) {  
        currentKondisi = "BOCOR";  
        thing.stream(thing["Kondisi"]);  
        currentLokasi = "AREA 2";  
        thing.stream(thing["Lokasi"]);  
  
        if (currentTime - waktutelegram2 >= 300000) { // 300000 ms = 5 minutes  
            String pesan2 = "PERINGATAN! \n";  
            pesan2 += "Tanggal: " + String(Date) + "\n";  
            pesan2 += "Waktu: " + String(Time) + "\n";  
            pesan2 += "Kondisi Kebocoran: " + currentKondisi + "\n";  
            pesan2 += "Lokasi Kebocoran: " + currentLokasi + "\n";  
            myBot.sendMessage(CHAT_ID, pesan2, "");  
            waktutelegram2 = millis();  
        }  
    } else {  
        currentKondisi = "TIDAK BOCOR";  
        thing.stream(thing["Kondisi"]);  
        currentLokasi = "NULL";  
        thing.stream(thing["Lokasi"]);  
    }  
}
```